

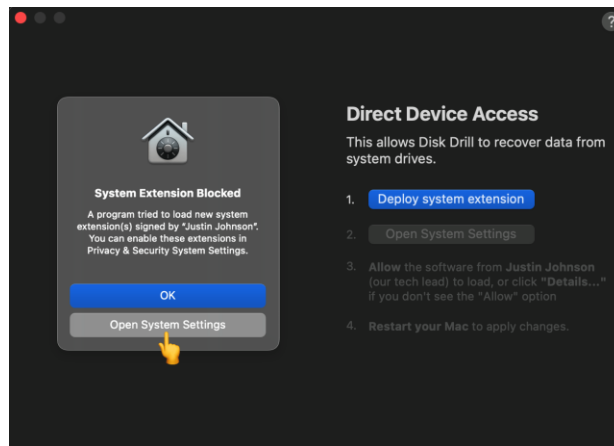
Jane Stashi  
CCleaner and Recuva Assignment  
Cyber Risk Management  
CIDM 6341

**Question 1: what did you do?**

**Provide a description of what you did. In particular, I want to know what you did to research the subject of the report. In most cases the value of the research is in the journey more than the result, same here, the better/clearer/insightful your write up is the better it scores.**

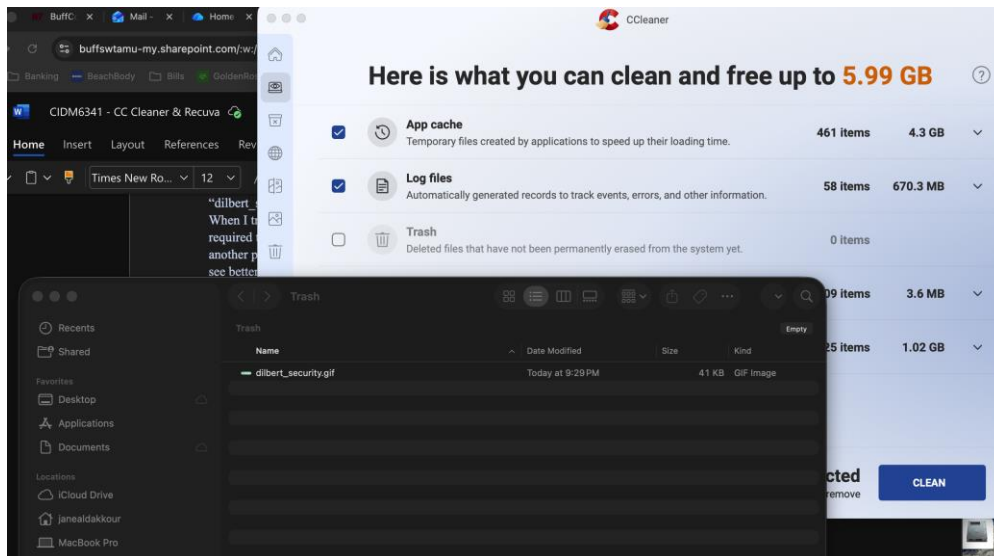
To start this assignment, I read through the provided instructions then went to the link within those instructions. The link took me to the Recuva download page. I downloaded the free version from this webpage but received an error when I attempted to open it. I then went back to the Recuva website and followed the instructions to find the Mac version. I then downloaded this version and was able to open Recuva. I then went back to the assignment instructions and downloaded the `dilbert_security.gif` file as instructed. I saved it to my desktop. I then continued to follow the instructions, which told me to throw away the `dilbert_security.gif` file. I first emptied the trash, then threw away the `dilbert_security.gif` file and recovered it from the trashcan (before emptying it). I then “recovered” the `dilbert_security.gif` file and dragged it from the trashcan back into the “CIDM6341 Cybersecurity” folder I have. I then checked that I was able to reopen the file, which I was. I then followed the next step, which was to move the file back to the trashcan and permanently delete the file.

When I went to the next step, I realized that the version of Recuva that I had downloaded was just the CC Cleaner, and not the Recuva document recovery. Upon doing more research, I discovered that there was not a version of Recuva that was developed for iOS, as it was only developed for Windows software. I then used Google’s AI, Gemini, to discover other programs that would allow me to recover deleted files, in which there were several that were listed. At first, I chose the software Disk Drill. I downloaded it and tried to have it run a deep scan of my computer for recovery; however, it was unable to do so as the permissions were being blocked. I selected the option in Disk Drill to show how to expand the search, which took me to an instruction video that explained how to lower the security settings on my Mac to allow for a deep scan to be performed. These instructions provided showed how to access the “start-up utility settings” on my Mac to reconfigure them to allow Disk Drill deeper access to my Mac’s hard drive (CleverFiles, 2023).



After following the video instructions, I was able to perform a deep scan of my computer's drive. I used the search bar to search for the "dilbert\_security.gif" image and came back with no results. I then filtered the found files by ".gif" to locate the one that I was looking for. There were many ".gif" files that were found, however I was unable to find the "dilbert\_security.gif" file that I was looking for. When I searched for it, no results came up. When I tried to recover a ".gif" file that was a similar size to the one I was looking for, it required that I purchase the "premium" version to do so. It was then that I decided to try another program called "EaseUS Data Recovery Wizard". With this software, I was able to see better previews of the images for recovery, however I received the same results from the Disk Drill scan. My next step was to research the reason behind the failed scans.

For the next part of the assignment, I was to once again download and delete the dilbert\_security.gif file, which I did. Once that was done, I was supposed to use CCleaner to clear my trash bin, which I did. I did this by opening CCleaner and went to the "Quick Clean" section, where there were various options for items that could be cleaned, one of which was the "trash" section. The trash section was greyed out so that it could not be selected, even though I had an item – the "dilbert\_security.gif" in the trash can that was not completely deleted.

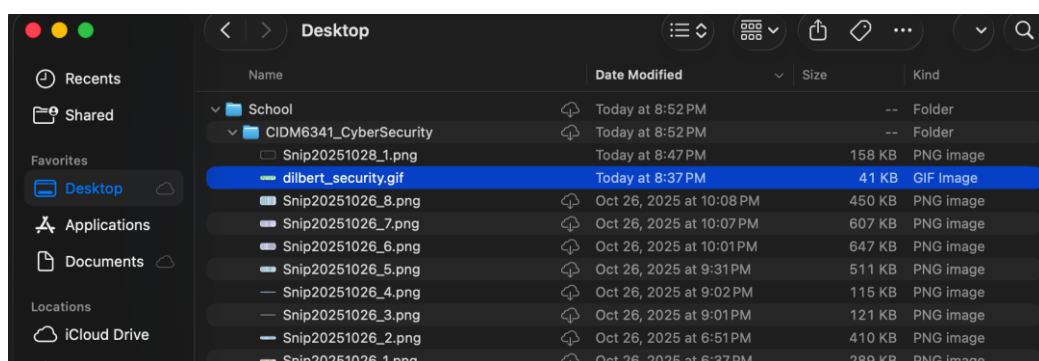


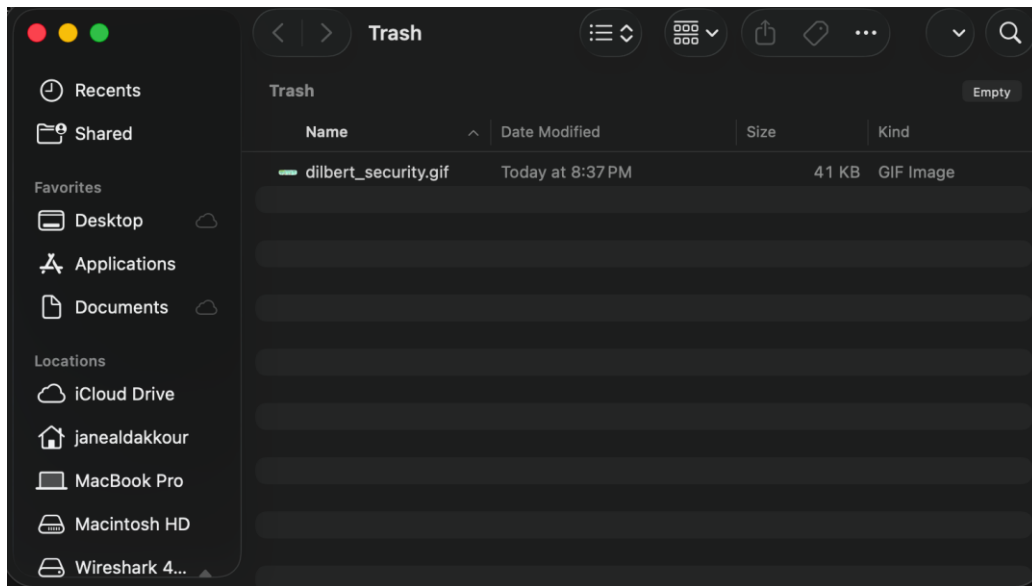
I scanned through some of the other items that CCleaner was recommended be cleaned. Since CCleaner was not picking up the `dilbert_security.gif` file in the scan that was performed, I went ahead and fully deleted it from the trash. I anticipated that the same results as the original deletion and recovery, which is that the file would not be found by Disk Drill.

## Question 2: what were the results?

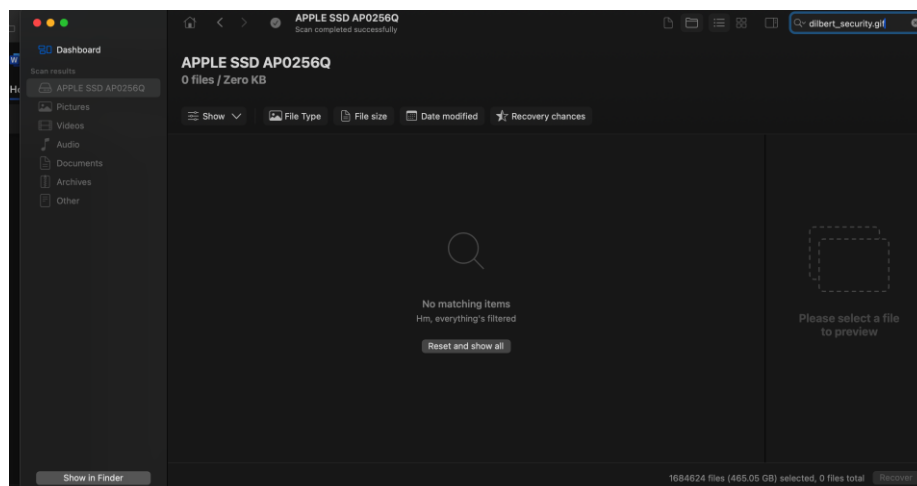
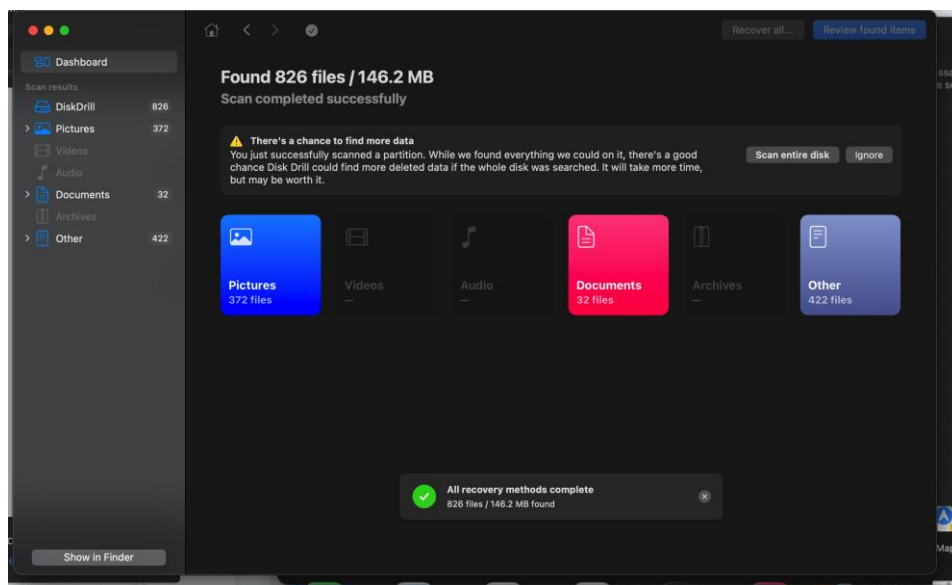
**Summarize and explain what your findings mean. Provide any printouts of products produced, this could be a report, a map, a table, etc.**

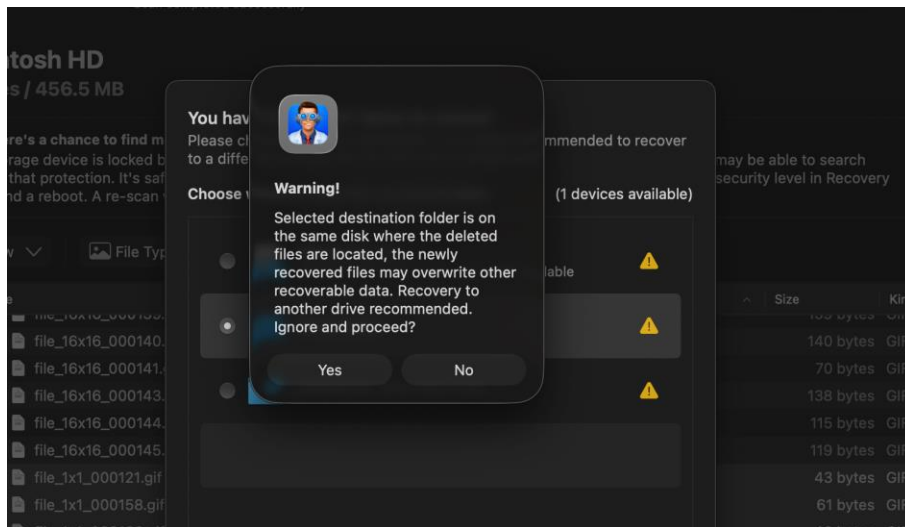
The images below are from saving and deleting the “`dilbert_security.gif`”:



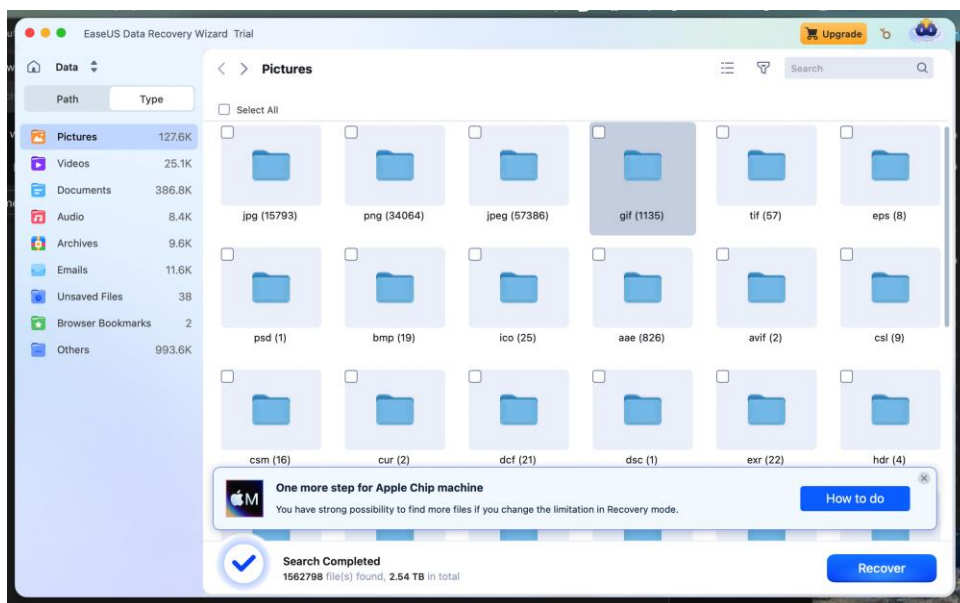


Images from the Disk Drill attempt:





Images from the EaseUS Data Recovery Wizard attempt:



The results of these attempts at file recovery were that I could not locate the file `dilbert_security.gif` file that I was attempting to recover.

**Question 3: what did you learn?**

**I can't tell you what you learned. What I will say is that I reward insight. Insights are aha moments (a term in use long before Oprah wanted to copyright it). If you see new ways of doing things, new insights to your thought processes, potential future applications be they personal or work related, crossovers to other topics, these are what I reward more than just telling me you learned lots. I expect you to learn lots, but it**

**isn't till you explain where and what that I see that you really did. Ok, so sometimes you don't learn much. I'll still grade this area high if you tell me why, what you know, how this works on what you've done in the past, etc. Sometimes when you start doing this you see that what you've learned is reinforcing what you've done and sometimes you even have small aha moments. Bottom line is to be reflective, think a few minutes or overnight about what you've done and how it fits into your nomological net (your personal set of knowledge base structure, those theories and beliefs that guide how you evaluate and use knowledge). Then write the section, when I see this done, I always score the section higher.**

My failed attempts at recovering the `dilbert_security.gif` file taught me much about the deletion and file recovery process, as I researched about different operating systems procedures and processes for treating deleted data. I have a Mac which operates on the Mac operating system. This operating system treats deleted data differently from different from the Windows operating system. When a file is deleted on the MacOS, it first goes into the “Trash” section, which is a “temporary holding place” for deleted files (Darwin’s Data, n.d.). This trash bin will hold deleted files until the empty button is selected or until 30 days passes, and the trash file allows for deleted files to still be recovered within the time that they are still in there. Once the data is deleted from the trash bin, however, the data is no longer recoverable. It is at that point that the data is permanently deleted and becomes written over by new data making it unrecoverable. This process is more streamlined for data processing; however, it makes it so that the data is not recoverable, which explains why on my Mac I was not able to find the “`dilbert_security.gif`” file using the Disk Drill deep scan. My Mac likely would have written over the deleted data, which explains why that file could not be found or recovered. The reason that MacOS does this is because MacOS has a solid-state drive (SSD), which is a newer more advanced technology that uses “flash memory” to store data rather than spinning disk (GeeksforGeeks, 2025). This makes it much faster and more efficient than using a hard disk drive (HDD), which is the other, more traditional type of memory. The HDD has spinning disk onto which data is read and/or written (GeeksforGeeks, 2025). The SDD works with Apple software to run a protocol called “TRIM active” which “erases the directory area” and designates each of the “data drives for re-use” (merrillhou, 2021). This means that it also “over-writes every superblock that was previously in use to prepare it for re-writing” meaning that the data is overwritten and unrecoverable (merrillhou, 2021). With HDD, just because a file is deleted does not mean that it has been destroyed, erased, or overwritten by new code on a drive. The means that the data is still recoverable from the HDD itself, as it is physically written onto the drive and until it is written over, will remain there.

## References:

[1] CleverFiles. (2023, September 21). How to Enable Disk Drill's System Extension

(macOS Ventura, Sonoma) [Video]. YouTube.

[https://www.youtube.com/watch?v=ioAVoX\\_3Aic](https://www.youtube.com/watch?v=ioAVoX_3Aic)

[2] Darwin's Data. (n.d.). Are files permanently deleted from Mac?

[https://darwinsdata.com/are-files-permanently-deleted-from-mac/#google\\_vignette](https://darwinsdata.com/are-files-permanently-deleted-from-mac/#google_vignette)

[3] GeeksforGeeks. (2025, July 11). Difference between Hard Disk Drive (HDD) and Solid

State Drive (SSD). <https://www.geeksforgeeks.org/computer-organization-architecture/difference-between-hard-disk-drive-hdd-and-solid-state-drive-ssd/>

[4] merrillhou. (2021, November 10). is it possible to restore data after erasing with disk

utility [Online forum post]. Apple Support Communities.

<https://discussions.apple.com/thread/253350992?sortBy=rank>